

INTRODUCTION

43 years after the Mianus River Bridge collapse stole the lives of 3 motorists due to undetected corrosion, inspectors still rely on the same visual methods that led to the accident (NTSB, 1984). **Traditional methods for inspecting infrastructure miss 96% of crack indications** (Phares et al., 2001), and require snooper trucks **costing \$4,600+ per bridge** (Federal Highway Administration, 2019). Such methods also **produce fall hazards for workers**, and render thousands of 2D images without meaningful context, causing inspectors to be unable to localize defects on the structure. Today, **42,400 U.S. bridges are rated in poor condition**, as the **rehabilitation backlog has reached \$191.3 billion, growing faster than repair capacity** (ASCE, 2025).

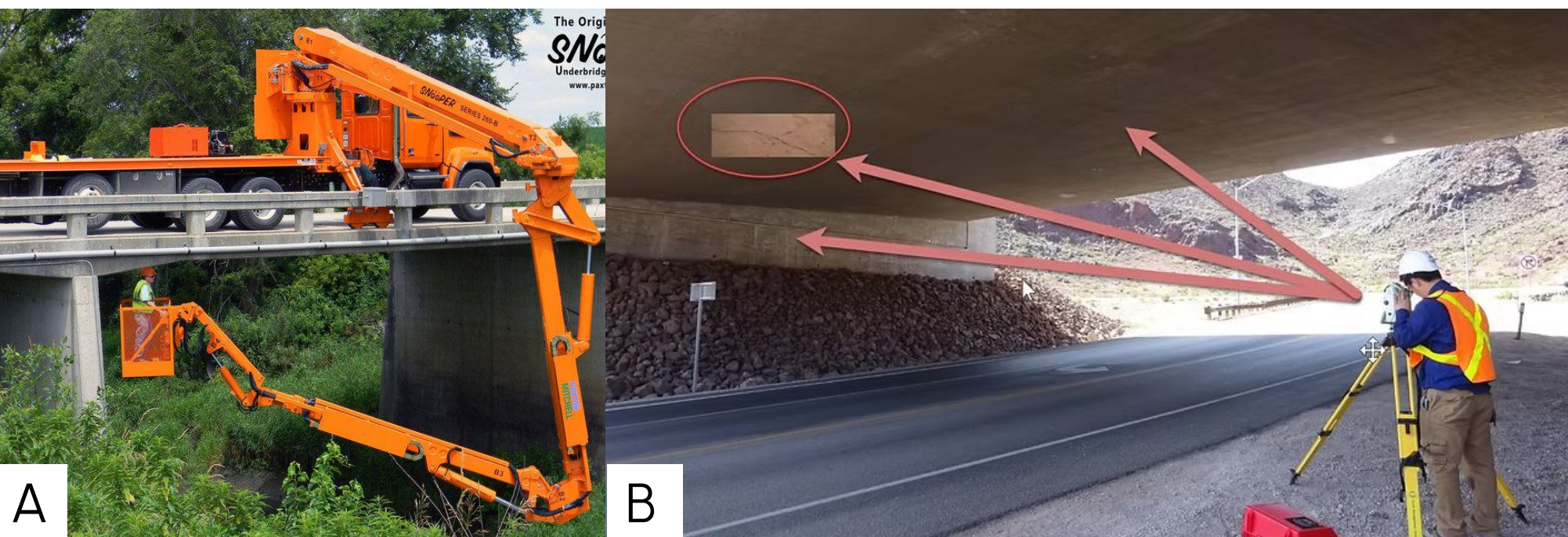


Figure 1. Image of snooper truck (A) (Core77, 2024) & methods of visual inspection (B) (Infrastructure Preservation Corporation, 2017). These traditional methods of infrastructure inspection are both hazardous (A) and inefficient and unreliable (B's use of human visual inspection is unreliable and takes training and discernment that do not produce consistent, accurate results). Diagram created by Finalists in Google Slides, images from aforementioned sources.

Several existing technologies prove to be more efficient solutions for inspection yet are not unified in implementation. UAVs have cut bridge inspection costs by 75% and time by 70% (FHWA, 2019), removing workers from hazardous scenarios while accessing structural areas that equipment may not be able to reach. Deep learning models such as YOLOv11 can detect and outline defects at pixel accuracy in real time, enabling automated diagnosis and efficient surveys. 3D Gaussian Splatting (3DGS) is a new technology capable of high-quality, photorealistic 3D reconstruction from overlapping 2D imagery orbiting a subject (Kerbl et al., 2023). As compared to classic 2D inspection images, 3D models grant inspectors spatial context that allows damage to be more easily located on the structure. Given all these technologies, what does not exist yet is a cohesive platform. Detection systems produce isolated 2D annotations without spatial localization. 3D reconstruction technologies can create robust models without diagnostic reporting. Detection and mapping have not yet been unified and **3D Gaussian Splatting has not been applied to defect visualization in infrastructure inspection**.

RESEARCH OBJECTIVES

Engineering Problem: Infrastructure inspection is largely manual, and technologies to address inefficient inspection workflows are mature but fragmented. Detection models, 3D reconstruction, and autonomous flight exist independently but are viable for making inspection more efficient.

Research Question: Can a combined pipeline including machine learning inference, autonomous flight control, and 3D Gaussian Splatting produce actionable spatial inspection through a defect map of infrastructure?

GOAL: Develop a custom pipeline to collect infrastructure imagery, accurately detect structural defects, and localize them onto a precise 3D model of the scanned structure.

Engineering Design Criteria

Robustness	Functionality	Specification
Detection generalizes across three defect classes: cracks, corrosion, and irregular spalling	Survey → detect → localize → visualize executes as one deployable sequence with little human input	~50% lower build cost, onboard ML inference > 30 FPS (real-time), detection Mean Average Precision > 0.50

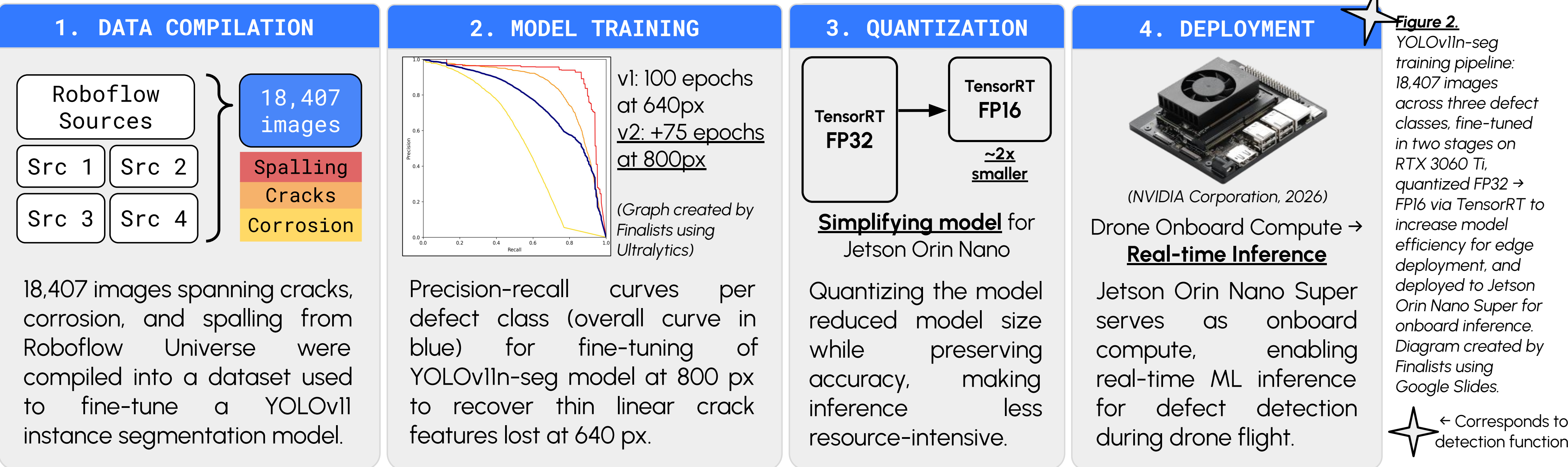
Desired Project Outcomes

- Prototype a powerful vision quadcopter for **real-time defect detection**
- Detect 3 defect** classes (cracks, corrosion, spalling) at high accuracy in real time
- Use **3D Gaussian Splatting** to produce a navigable, interactive model for analysis
- Localize detected defects in 3D space** on a reconstructed 3D Gaussian Splatting model

Adaptive Infrastructure UAV: Real-Time Defect Segmentation and 3D Gaussian Splatting for Enhanced Infrastructure Inspection and Diagnosis

ENGINEERING METHODOLOGIES

DEFECT DETECTION MODEL: A YOLOv11 Instance Segmentation model was fine-tuned to detect structural defects



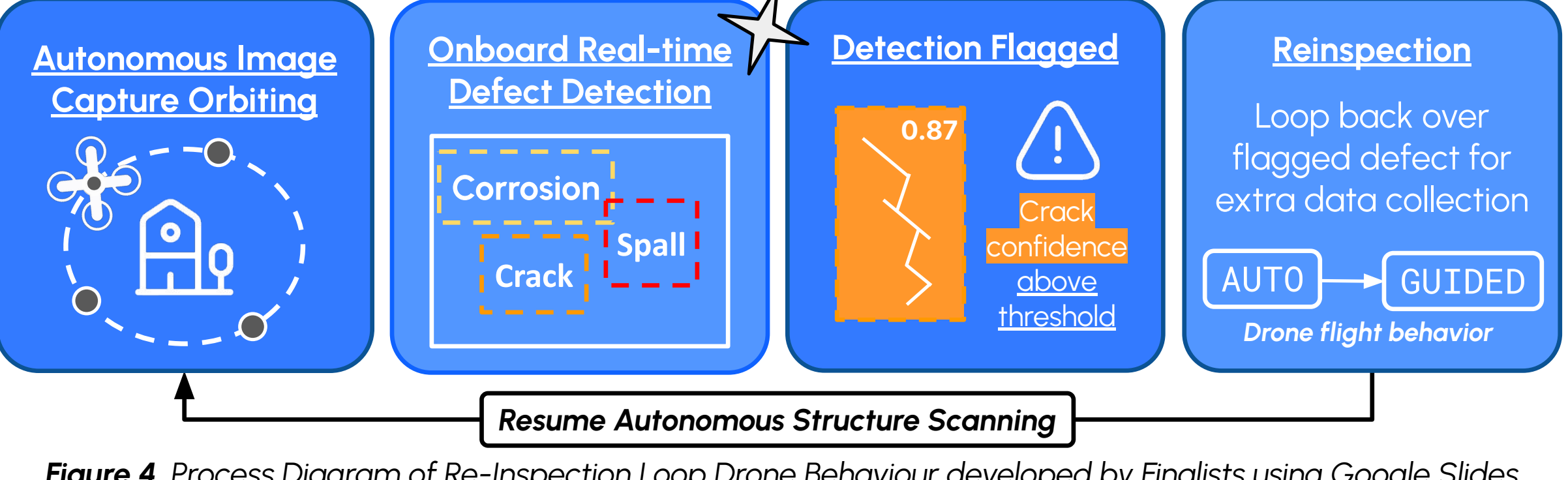
CUSTOM-BUILT UAV: Autonomous quadcopter engineered for real-time ML → structure data collection

Total Cost	\$1,041.22
Flight Time	~15 minutes
Inference Speed	120-160 FPS
Thrust-to-Weight	2.3:1

Figure 3. Prototyped Quadcopter (Left Image taken by Finalists) and **Table 1. Relevant Drone Specifications** displaying performance of drone. Table created by Finalists in Google Slides.

The quadcopter's onboard compute, the NVIDIA Jetson Orin Nano Super, alongside a 4K camera and SAHL, acts as a compute-and-capture mechanism, with the Pixhawk receiving MAVLink commands from the Jetson when the CLAR function triggers re-inspection. Onboard inference on the live camera feed enables autonomous mid-flight defect detection, with the 4K resolution preserving sufficient detail in order to flag damage such as cracks.

CLOSED-LOOP ADAPTIVE RE-INSPECTION: Detections trigger re-pass → extra footage → **better defect 3D modeling**



CLAR runs onboard YOLOv11-seg continuously during flight, producing class, confidence, and GPS-tagged location for each frame. A re-inspection triggers when two conditions are met: detection confidence exceeds 0.70, and the site is more than 3 meters from any previously visited location. On trigger, MAVLink commands the Pixhawk from AUTO to GUIDED, routes the drone to the flagged coordinate for additional multi-view capture, appends the site to the visited list, and returns control to AUTO. The waypoint mission resumes from its last position.

DEFECT MAPPING PIPELINE: Map the detected defects & develop 3D model to create novel 3D visualization

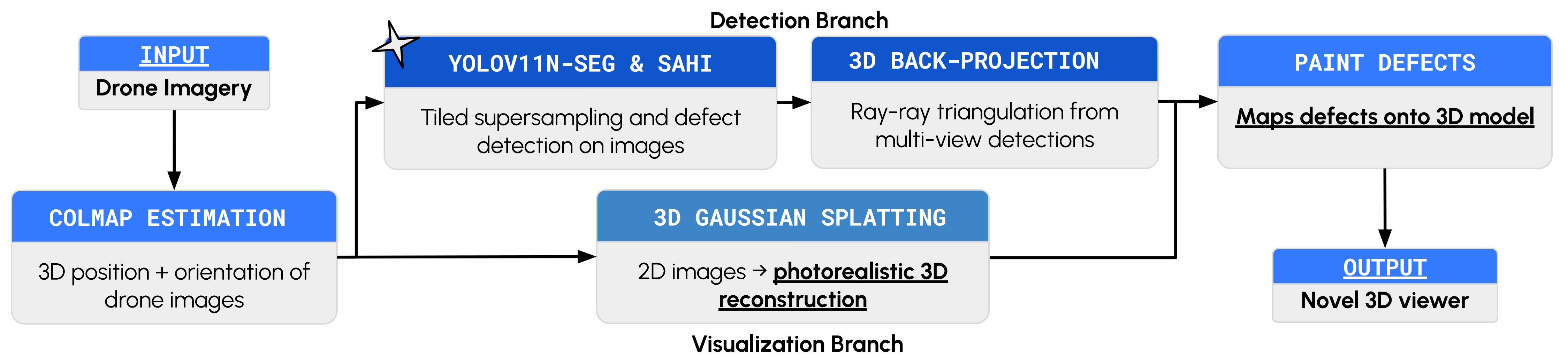


Figure 5. Pipeline demonstrating process of ingesting images to output combined 3D Gaussian Splatting model with detected structural defects painted on. Created by Finalists using Google Slides.

Drone imagery is processed through a two-branch pipeline. COLMAP estimates per-frame camera poses and a sparse point cloud that seeds both branches. The detection branch runs SAHI on the original images that are then fed into the YOLOv11-seg model, and back-projects detections into 3D space via ray-ray triangulation, with DBSCAN clustering consolidating redundant views. The visualization branch trains a 3D Gaussian Splatting structure model. The branches merge at the painting step, where KDTree ray-marching tags and tints Gaussians by class. The output is a navigable 3D viewer with spatially-anchored defect labels.

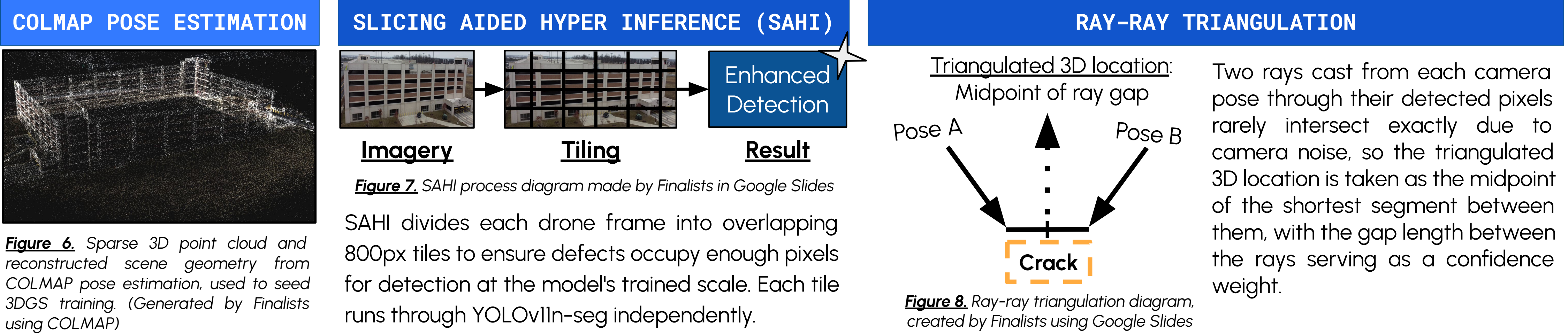


Figure 6. Sparse 3D point cloud and reconstructed scene geometry from COLMAP pose estimation, used to seed 3DGS training. (Generated by Finalists using COLMAP) **Figure 7.** SAHI process diagram made by Finalists in Google Slides **Figure 8.** Ray-ray triangulation diagram, created by Finalists using Google Slides

RESULTS

CLASS	mAP50	RECALL	PRECISION
Spalling	0.947	0.899	0.892
Cracks	0.822	0.750	0.788
Corrosion	0.518	0.460	0.645
Overall	0.762	0.703	0.775

Figure 9. YOLOv11-seg model metrics (Chart created by Finalists using Google Slides)

Setting mode to GUIDED...
COMMAND_ACK {command : 176, result : 0}
HEARTBEAT {type : 2, autopilot : 3, base_mode : GUIDED}

Figure 10. Successful CLAR Trigger Result

Terminal output during flight test demonstrating CLAR switching drone from AUTO surveillance to GUIDED re-passing behaviour. (Screenshot captured by Finalists)

FIELD TESTING SUMMARY

- Flight Time:** 10 minutes
- Images Captured:** ~3000
- Defects Flagged:** 8



Figure 12. Screenshot of 3D model prior to painting (created by Finalists via screenshot of nerfstudio WebGL viewer)

FINAL OUTPUT: Navigable 3D viewer with defects painted at exact locations → replacing 2D photos with 3D report



Figure 13. 3D Viewer of model painted & displaying found defects (Built by Finalists with COLMAP, nerfstudio, YOLOv11-seg, and custom algorithms.)

ANALYSIS

Detection Model Performance: The model achieved 0.762 overall mAP50 across three defect classes on a test dataset. Spalling reached the highest accuracy (0.947), likely since spalling produces large, geometrically distinct surface disruptions that contrast sharply against background surfaces. Corrosion was weakest (0.518); approximately 52% of corrosion cases were predicted as background, possibly indicating that diffuse rust patterns resemble weathered concrete too closely for reliable detection. Crack detection (mAP50 0.822) saw a substantial benefit from fine-tuning at the 800px resolution, which helped preserve linear features lost during the 640px training.

CLAR: Initial field testing demonstrated that single-pass flight surveys delivered insufficient coverage on smaller defects, in turn degrading the ray-ray triangulation accuracy during 3D back-projection. This inspired the Closed-Loop Adaptive Re-inspection (CLAR) behaviour, where detection events above specific confidence thresholds trigger re-passes at flagged defect sites. This adaptation based on limitations discovered during static survey flights is a refinement to the general inspection workflow rather than a feature specified at design time.

Two-Branch Architecture: The 2-branch design was validated through a key finding: defects visible in 4K imagery are unresolvable using 3DGS rendering. An earlier single-branch pipeline attempted detection on rendered 3DGS surfaces, which proved lossy at flight-distance fidelity. Fine surface features did not survive 3DGS reconstruction, and even super-resolution techniques could not recover details destroyed by rendering. Separating detection onto the original 4K imagery eliminated this loss entirely. Both the accuracy-critical and fidelity-critical tasks now receive inputs that are matched to their requirements, merging only at the defect-painting step to produce reliable detection with a coherent 3D output.

Field Deployment and Limitations: CLAR successfully triggered per-defect during testing, capturing additional imagery for denser defect coverage and improved triangulation. The full system was built for \$1,041.22, and pipeline post-processing runs in under an hour on an RTX3060Ti. Testing dataset metrics help to characterize model performance on the training distribution of imagery and do not directly establish field accuracy at novel sites. The parking garage deployment is a proof-of-concept demonstration, and verified accuracy claims for operational inspection workflows will require additional testing on structures with known defect data.

CONCLUSIONS

This system demonstrates that an autonomous UAV combined with 3D Gaussian Splatting can produce spatially accurate, navigable defect maps of infrastructure. The combination of real-time detection, adaptive re-inspection, and photorealistic 3D reconstruction addresses gaps that inspection workflows leave open, delivering capabilities not present in current systems.

Field Impacts:

- Replaces \$4,600+ snooper truck deployments with a \$1,041 system
- Eliminates fall hazard** for inspection workers
- 3D viewer replaces 2D photos with a navigable, color-coded inspection report → spatial context and coverage verification

Broader Impacts:

- Architecture is transferable to other inspection domains (pipelines, buildings, power infrastructure)
- Adaptive re-inspection loop generalizes to any drone-based ML detection task requiring targeted data collection**

Future work for this project would involve validation across structures with known defects to create field accuracy benchmarks.

CITATIONS

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